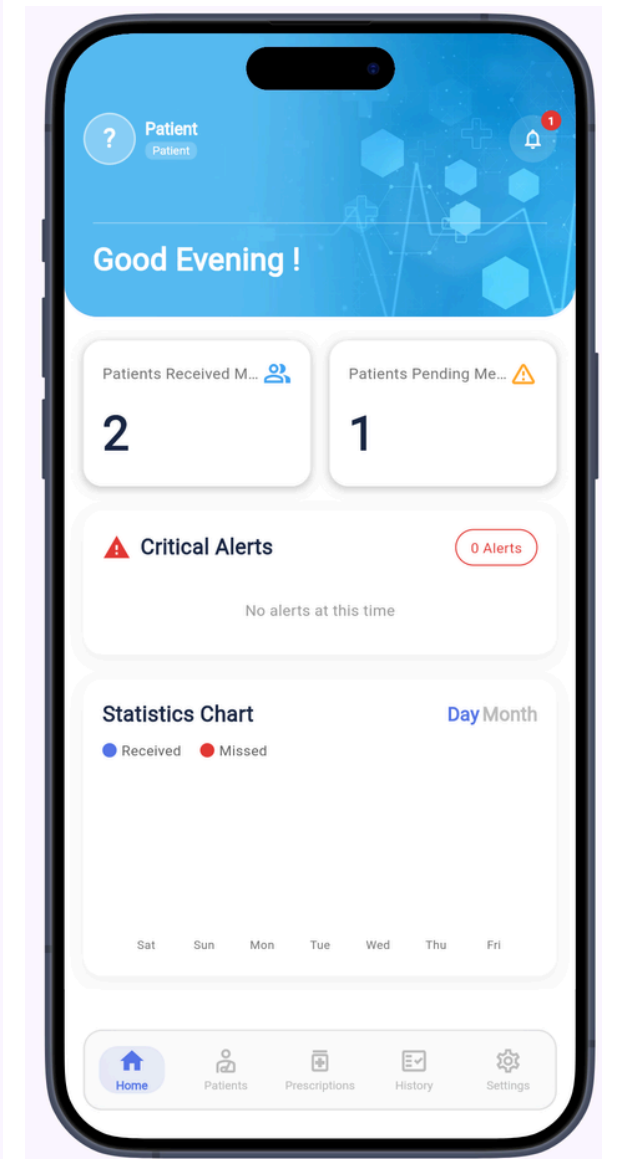
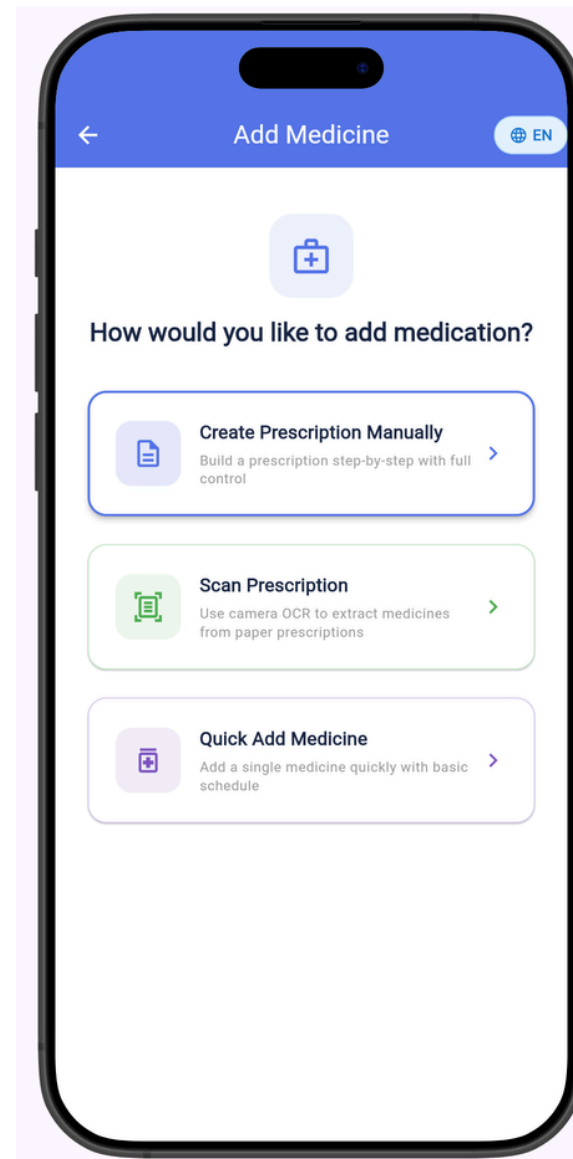




DasTern

Health Technology





Meet Our **DasTern** Team



Project Lead + OCR

Choeng Rayu



Frontend Developer

Sophal Taingchhay



Backend+Database

Lon Mengheng



AL-LLM-Service

Tet Elite



Frontend Developer

Tep Somnang



Frontend Developer

May Kunaphivath



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1. Introduction
2. Problem Statement
3. Propose Solution
4. Target Users & Needs
5. Core Feature & Functionality
6. Project Planning
7. Project Implementation
8. Challenges & Limitations
9. Demo
10. Conclusion
11. Evaluation & Future Work



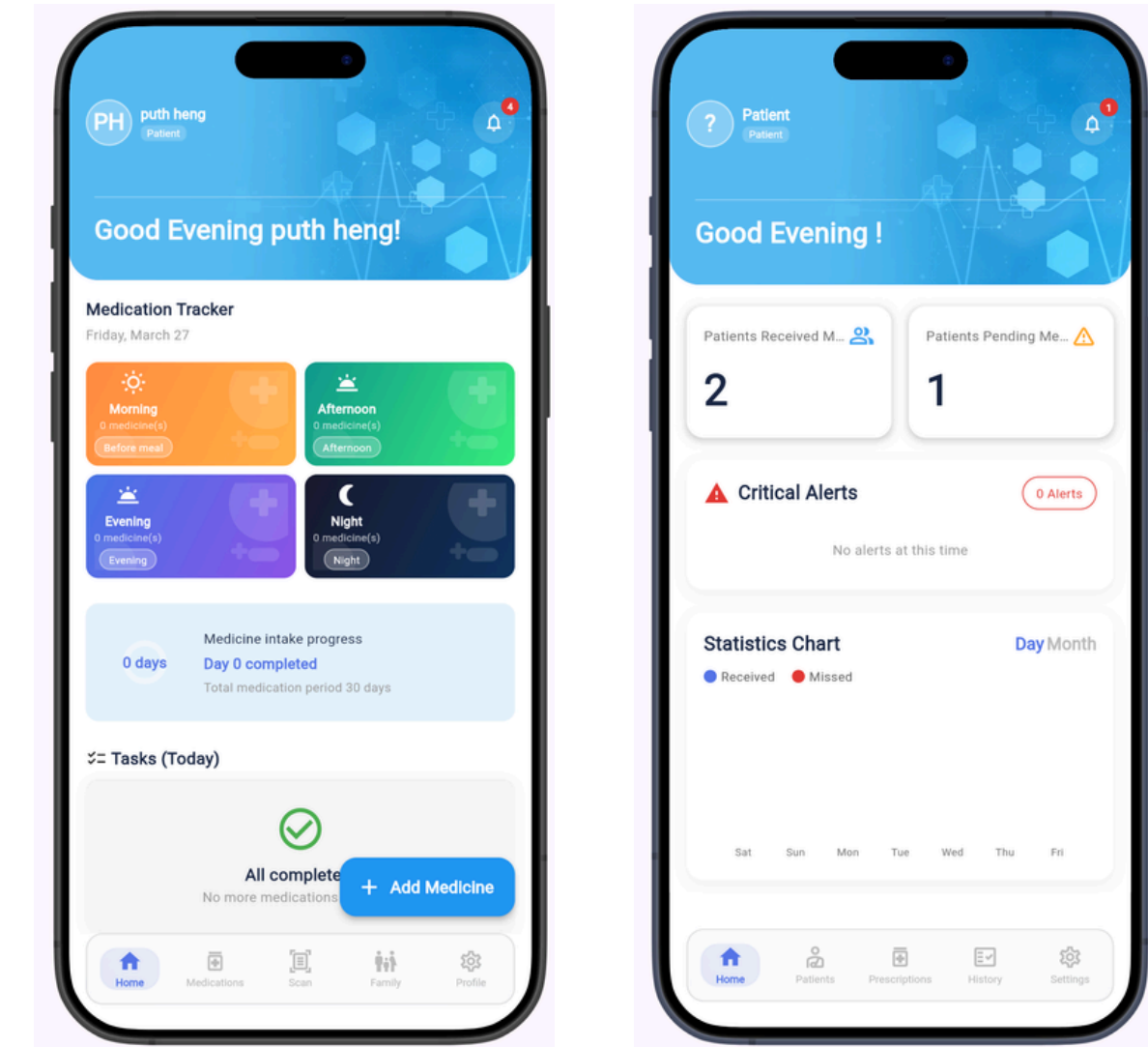
1. INTRODUCTION





Introduction

DasTern is a health-tech mobile application designed to improve **medication adherence** among Cambodian patients after hospital discharge. The application uses **OCR** technology to convert scanned paper prescriptions into digital medication schedules with automated **reminders**, family support features, and data insights for doctor analysis.





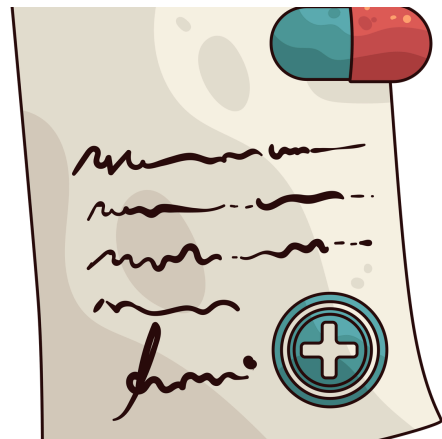
2. PROBLEM STATEMENT



2. PROBLEM STATEMENT



Paper-Based Prescriptions



Cambodian patients often use paper prescriptions.

Language Barriers



Instructions often in English or medical terms.

Limited Health Apps



Mobile health tools in Khmer are scarce.

Smart-Phone use



Mobile tech enables potential digital solutions.



3. PROPOSED SOLUTION



Proposed Solution

01



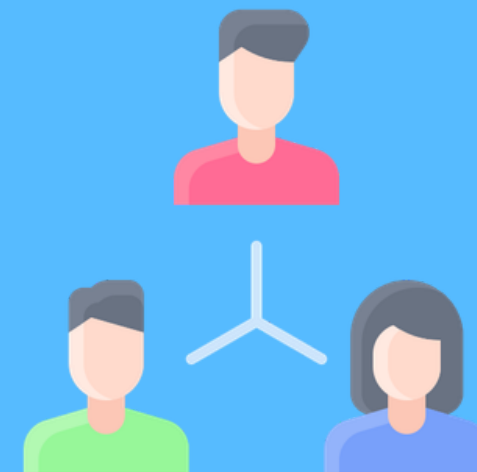
Offline medication reminders application

02



OCR-based **prescription scanning** for easy digitization of doctor's prescriptions

03



Family account linking to assist patients in following their medication schedule

04



Support doctors in **viewing patient records and prescriptions** for better follow-up care



Existing Solutions (MHealth Apps)

Application / System	OCR Prescription Scanning	Khmer Language	Offline Medication Reminders	Family / Caregiver Notification	Doctor Dashboard
 Medisafe	No	No	Limited	Limited	No
 MyTherapy	No	No	Limited	No	No
 carezone	No	No	No	No	No
	yes	Yes	Yes	yes	yes



4. TARGET USERS



Target User Groups

Patient



- **Elderly patients**
- Patients with chronic or post-hospital treatment
- Users with limited digital literacy
- Rural users with limited internet access

Family / Caregiver



- Relatives supporting elderly or dependent patients
- **Users responsible for monitoring medication adherence**

Doctor



- Medical professionals **prescribing and managing patient** treatment



5. CORE FEATURE & FUNCTIONALITY



PATIENTS FEATURES :



**Patient Prescription
Management**



Reminder System



**Dose Tracking &
Adherence**



**Data Ownership &
Security**



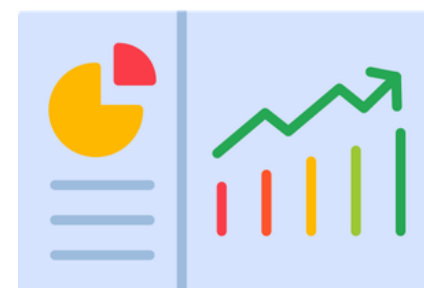
DOCTOR FEATURES :



**Doctor-Patient
Connection**



**Prescription
Creation**



Doctor Dashboard



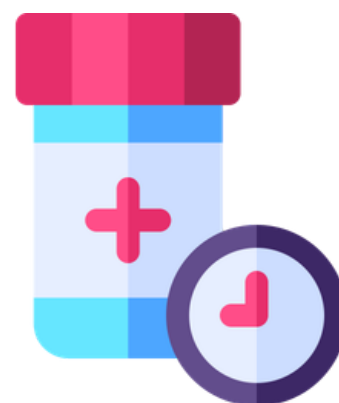
Patient Follow up



FAMILY & CAREGIVER FEATURES :



**Patient-Family
Connection**



**Alert Patient
Medication**



**Shared Monitor
Adherence**

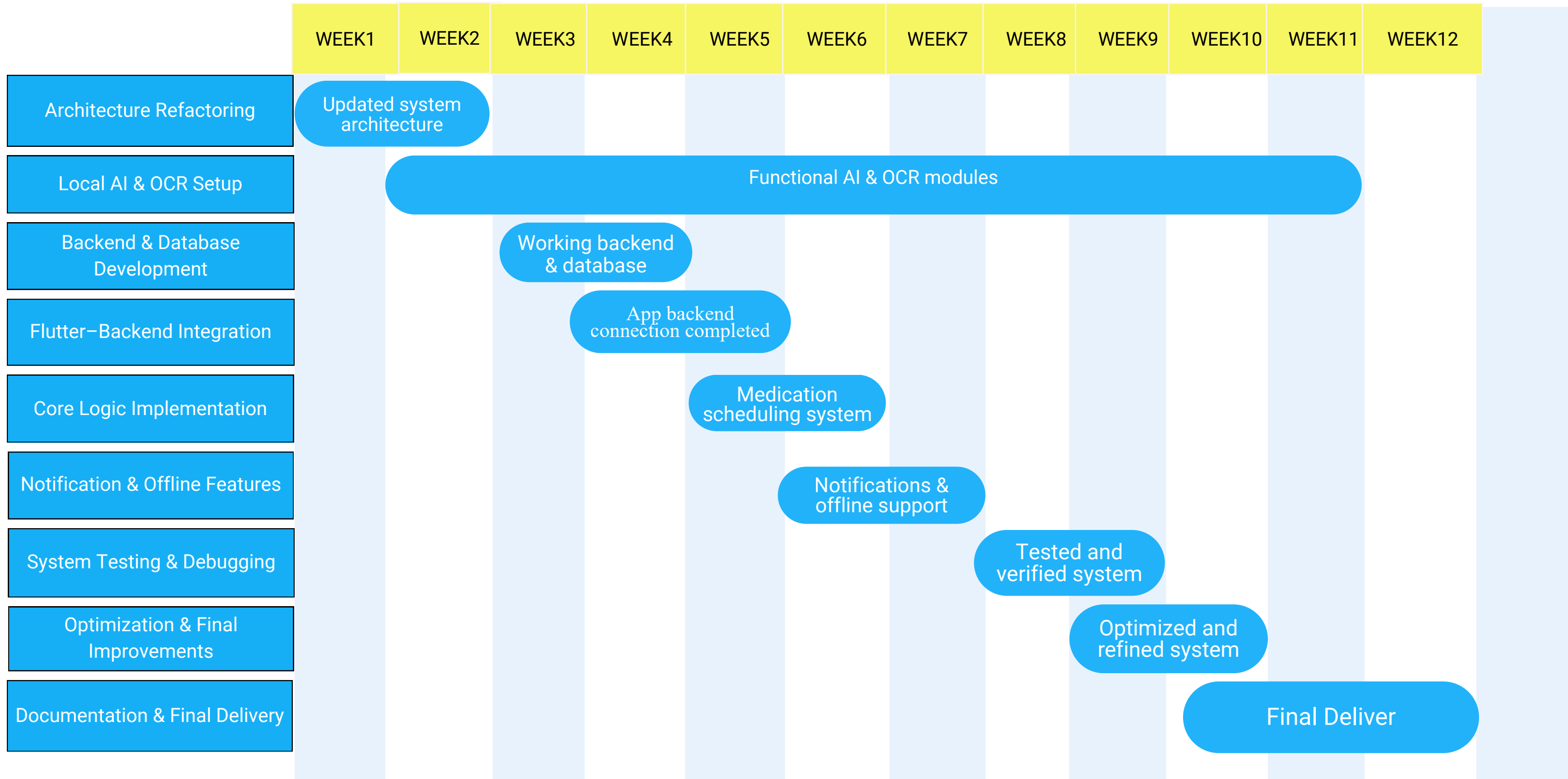


6. PROJECT PLANNING





Project Planning





7. PROJECT IMPLEMENTATION



Tools & Technology used

Design & Research



Collaboration



Telegram



Google Drive

AI Tools



MCP

LLaMA
by Meta



Ollama



Hugging Face

Development Technologies



MINIO

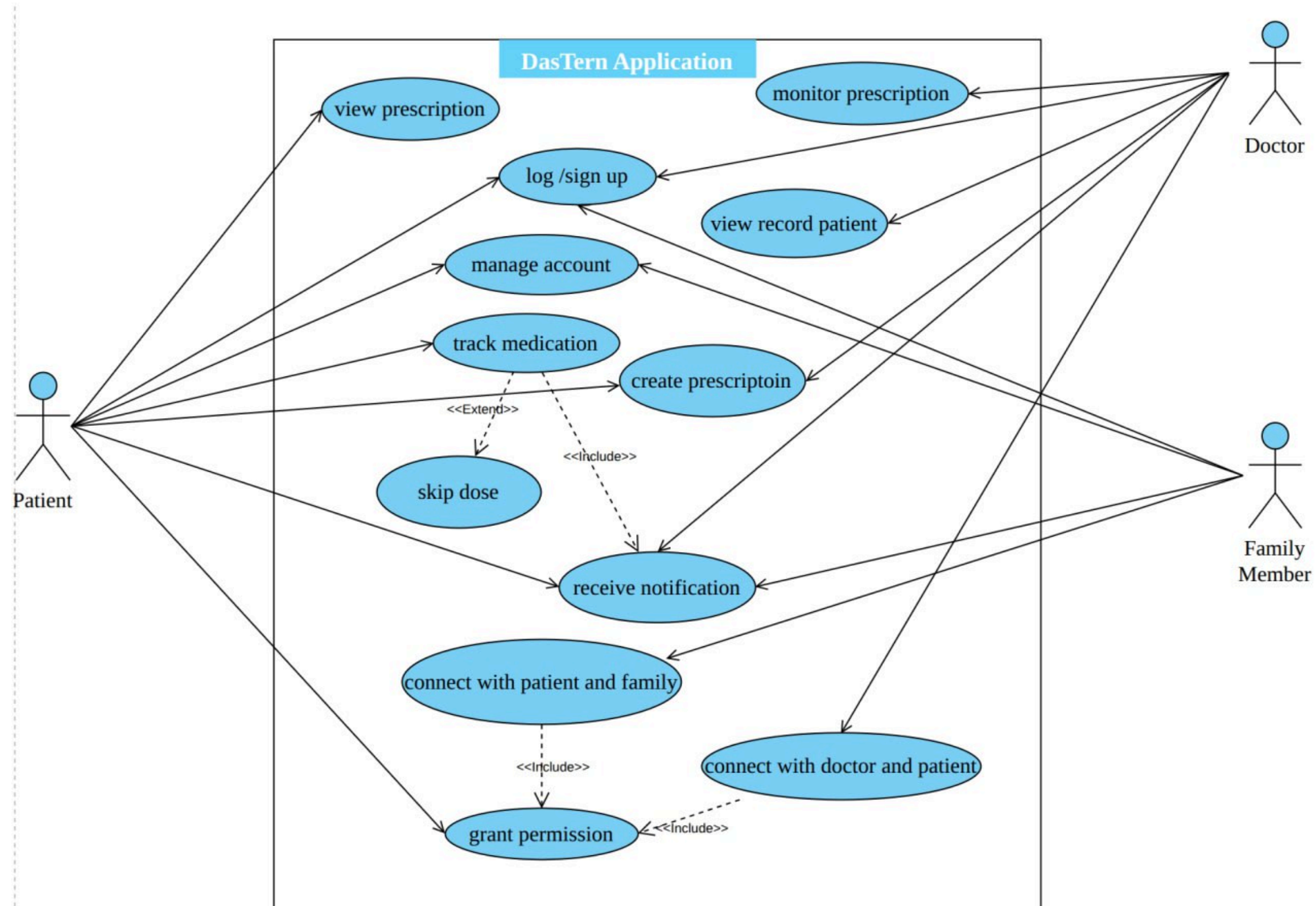


PostgreSQL

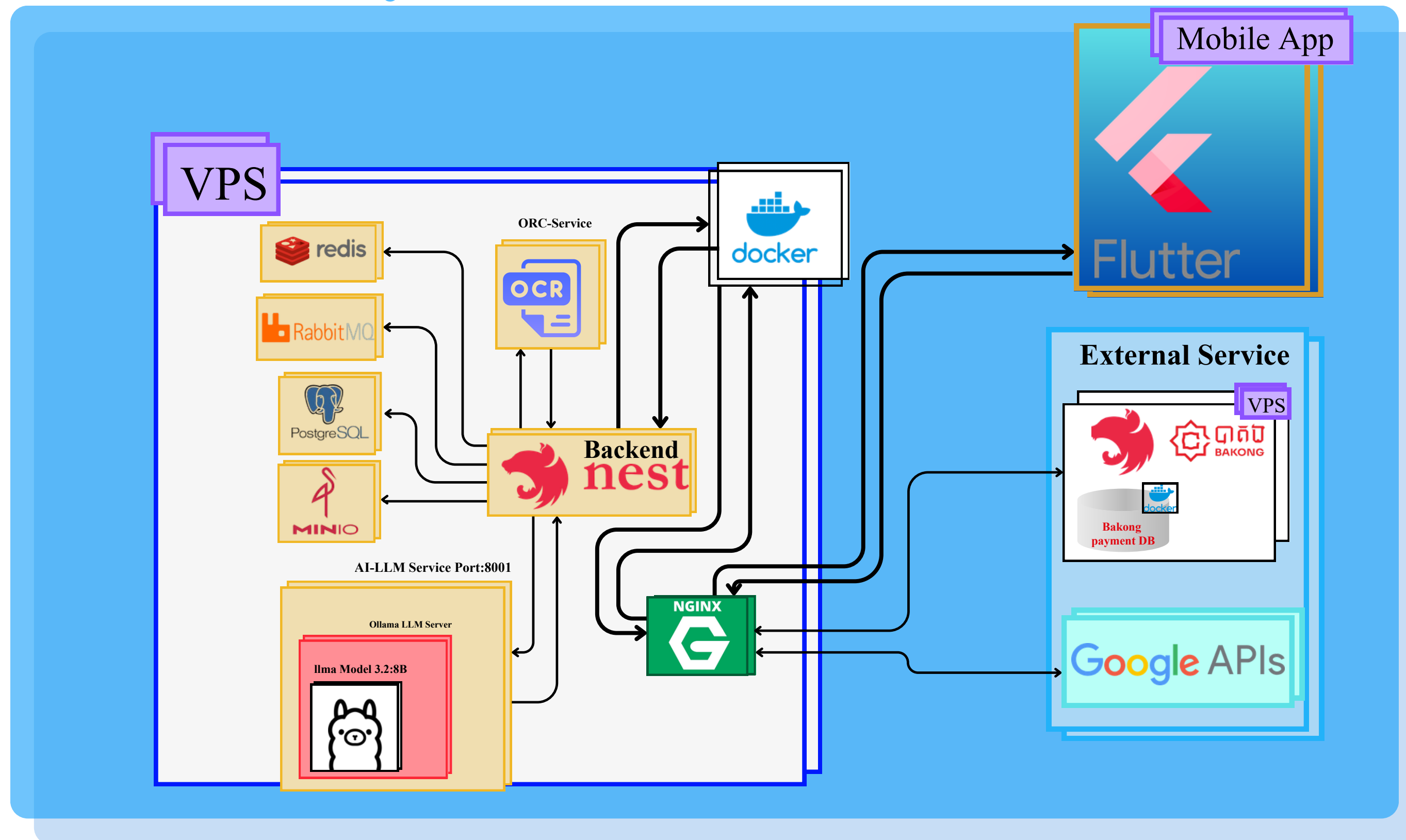


KIRI OCR
Lightweight English & Khmer
Document Recognition

Use Case Diagram

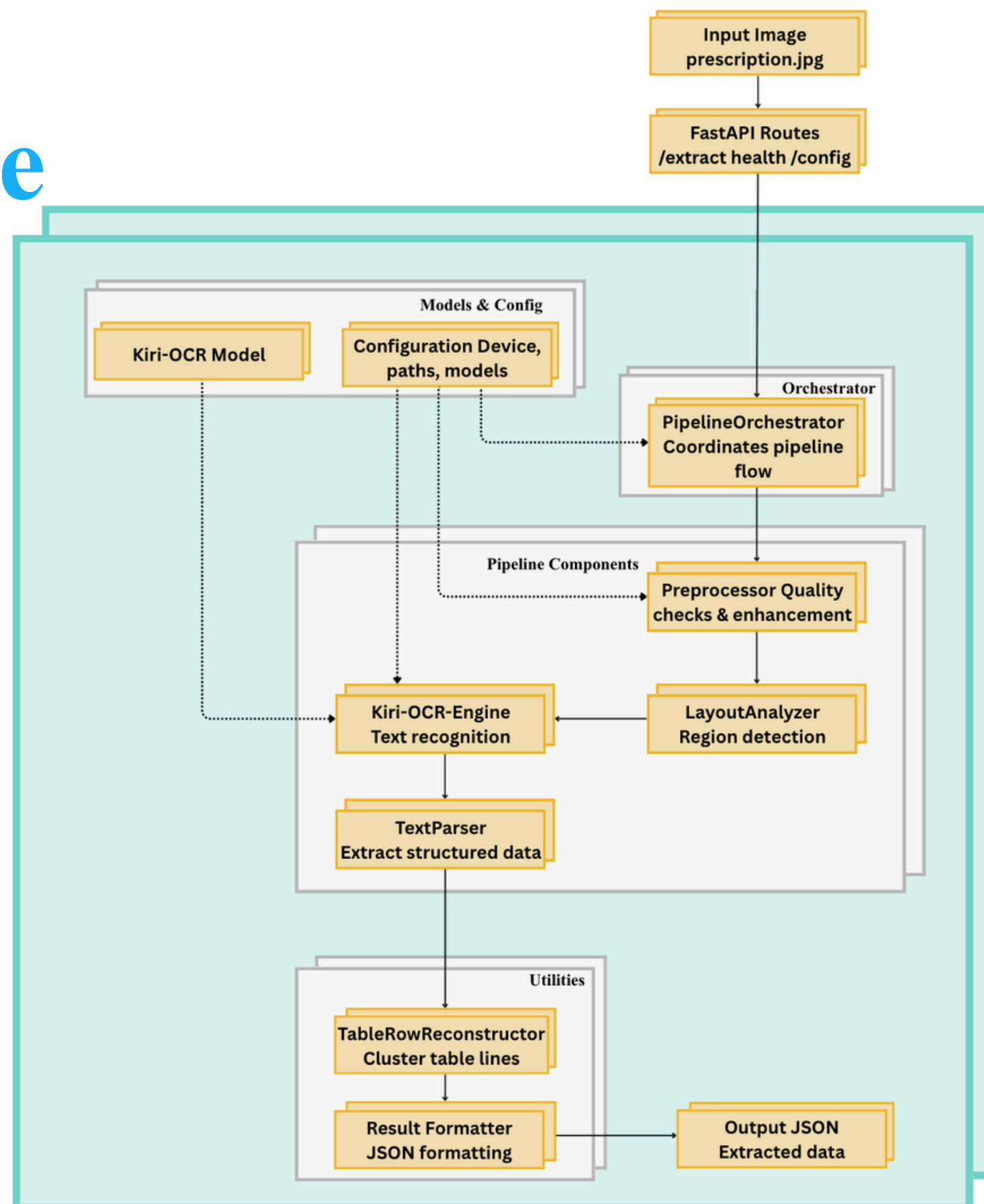


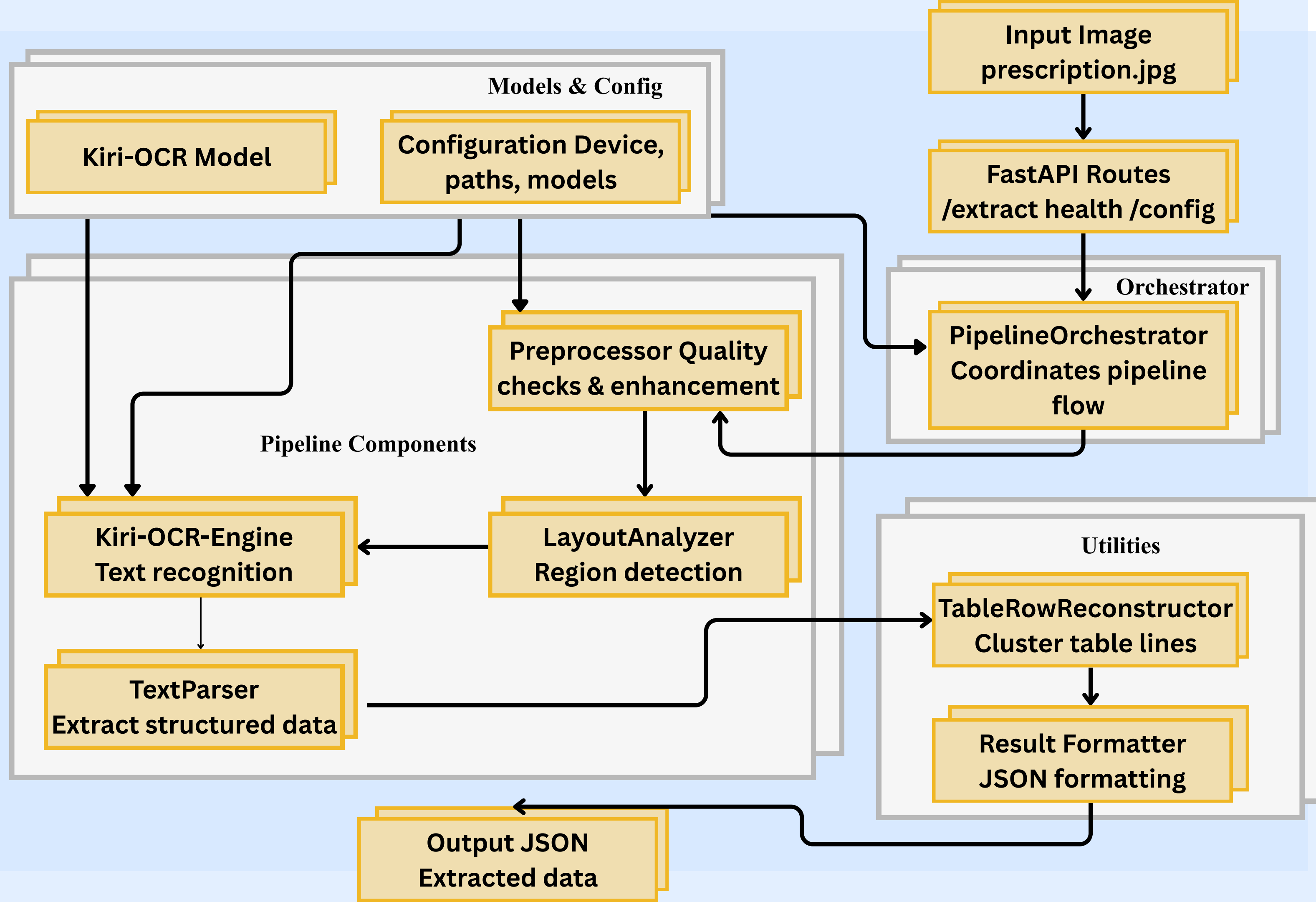
System Architecture



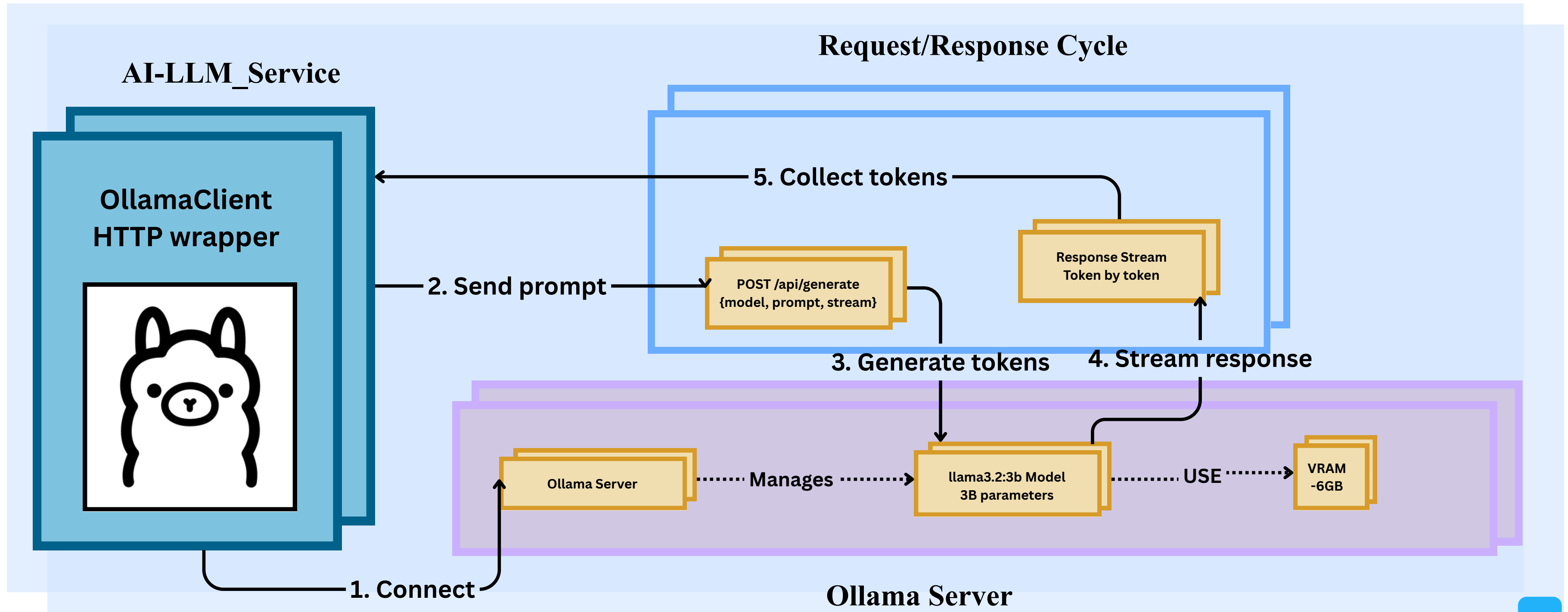


OCR Architecture





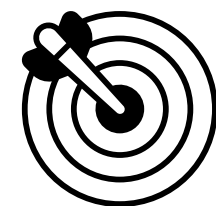
AI-LLM Architecture





8. CHALLENGES & LIMITATIONS



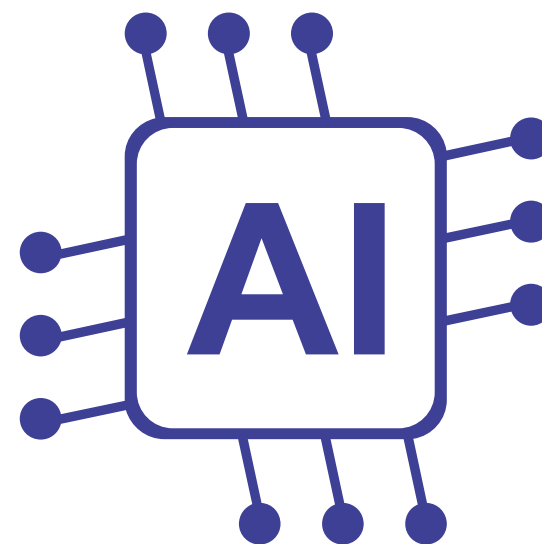


Challenges

Github conflict error



Running ai model locally

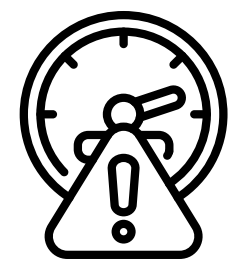


Finding Prescription for testing



Report Format Issue





Limitation

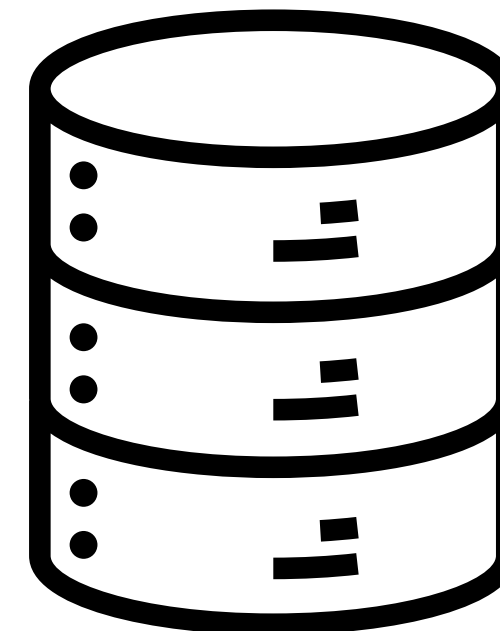
Handwritten prescription recognition



OCR result not yet 100% accurate



Local ai require large hardware support





9. DEMO





10. CONCLUSION





Conclusion

DasTern is now a fully functional mobile app that helps patients take their medicine correctly. By combining local OCR and smart AI, it instantly converts paper prescriptions into digital schedules, while the app support both Khmer and English and it is design specially for cambodia patients with voice reminders, and an offline mode, it provides a secure, reliable tool to prevent missed doses and ensure a safe recovery.



11. EVALUATION & FUTURE WORK



Future Work



OCR Feature

- OCR accuracy optimization
- Support French for OCR



Patient Monitor

- Provide health dashboard; blood pressure, temperature
- Adherence Analytics



Doctor Dashboard

- Doctor Dashboard Analytics
- Data Export & Reporting; patient record, analytic reports



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Thank You

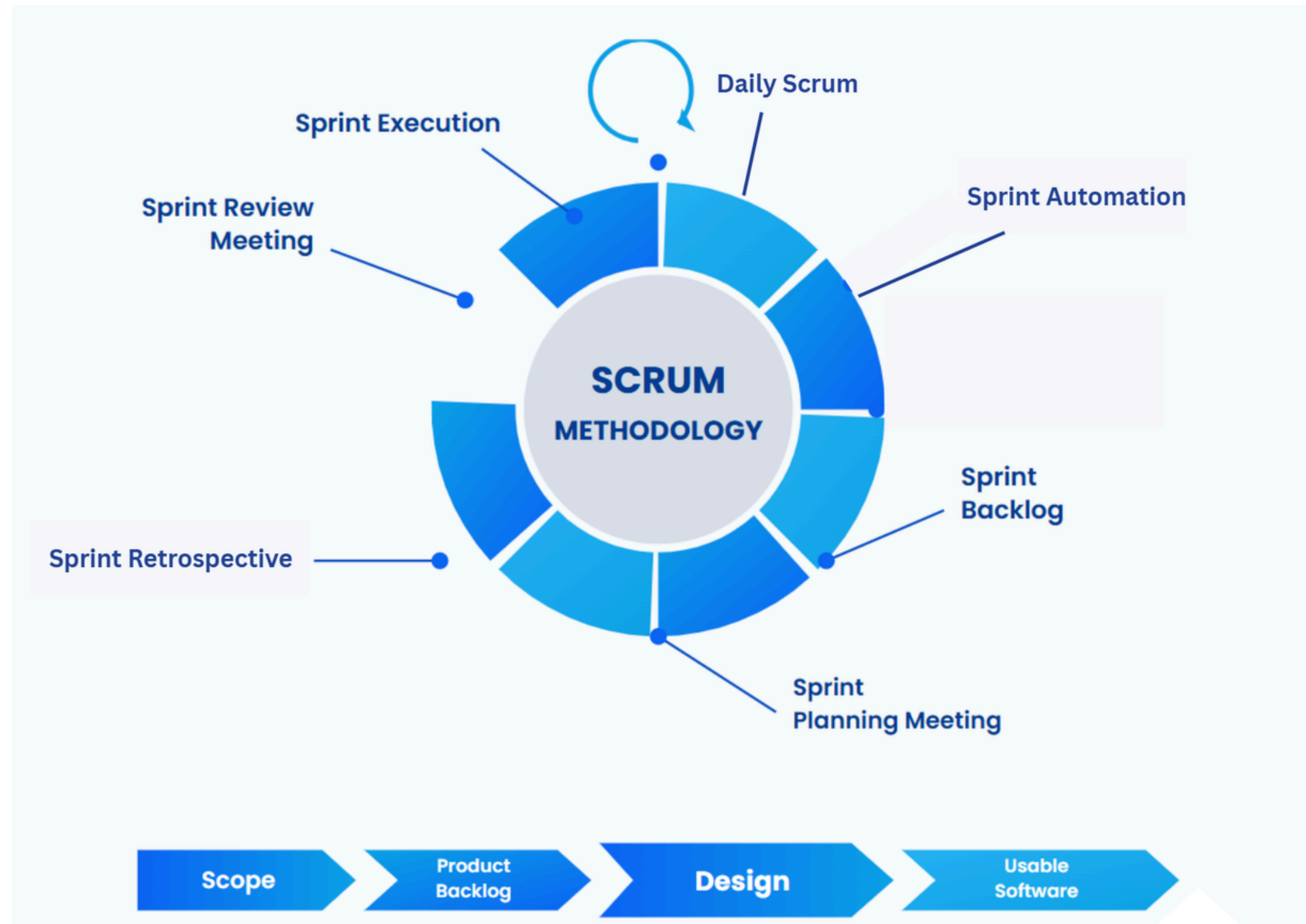




Appendices



Scrum Methodology





Task Breakdown

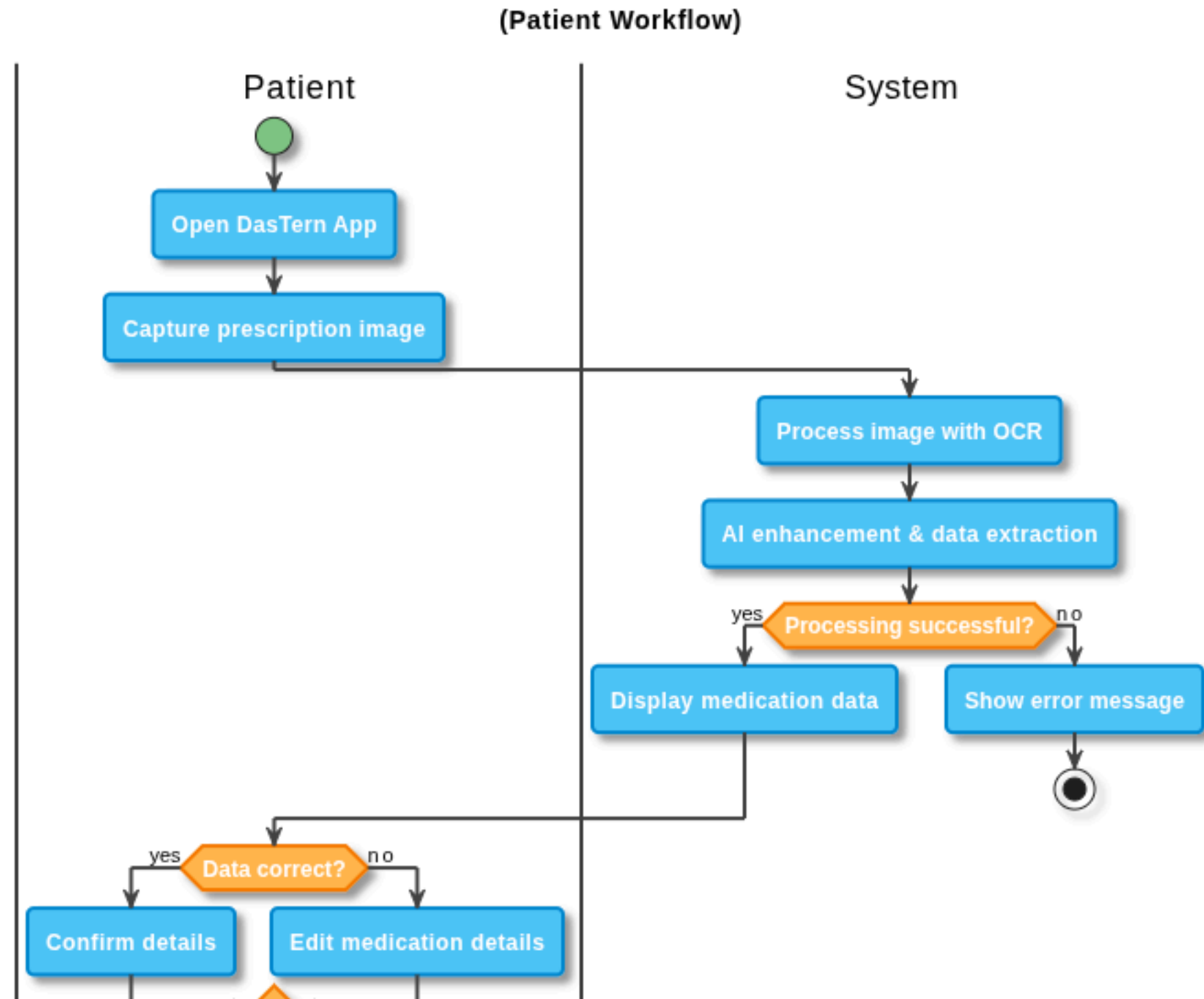


Member	Role	Main Deliverable	Status
Choeng Rayu	Project Leader & OCR Engineer	Project coordination, Local OCR implementation, OCR optimization & integration	Completed
Tep Somnang	Frontend Developer	Flutter UI implementation, API integration, Reminder logic	Completed
Tet Elite	AI LLM Engineer & UX/UI Support	LLM text enhancement (Ollama), Prompt engineering, JSON structuring, UI bug fixing	Completed
Lon Mengheng	Backend Developer	REST API development, Database implementation API testing	Completed
Sopha Taingchay	Frontend Developer	UI implementation, Feature integration, Validation handling	Completed
May Kunaphivath	Frontend Developer	UX/UI Design, Testing UI components, UI improvements	Completed

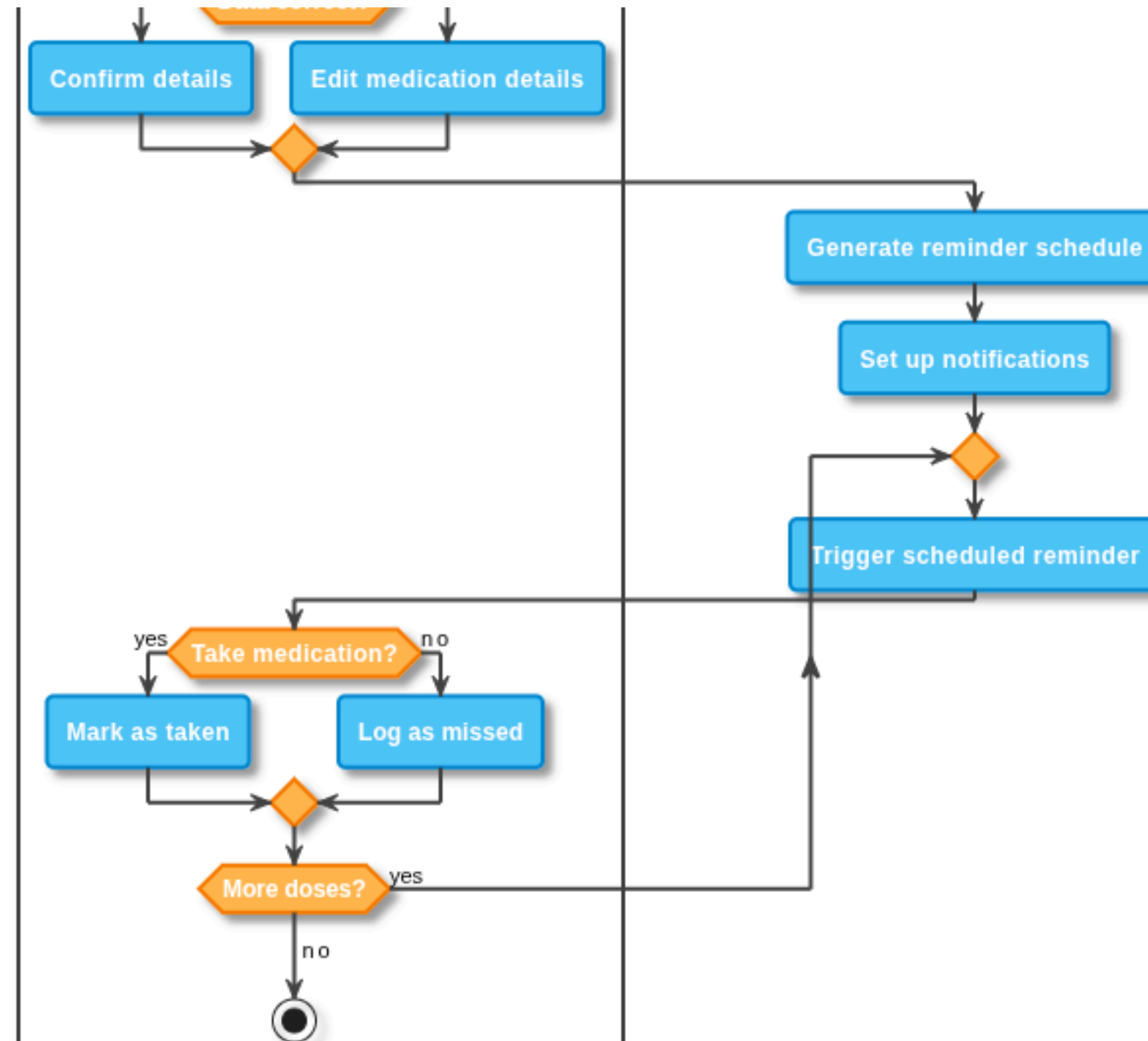


UML Diagram

Activity Diagram (Patient)

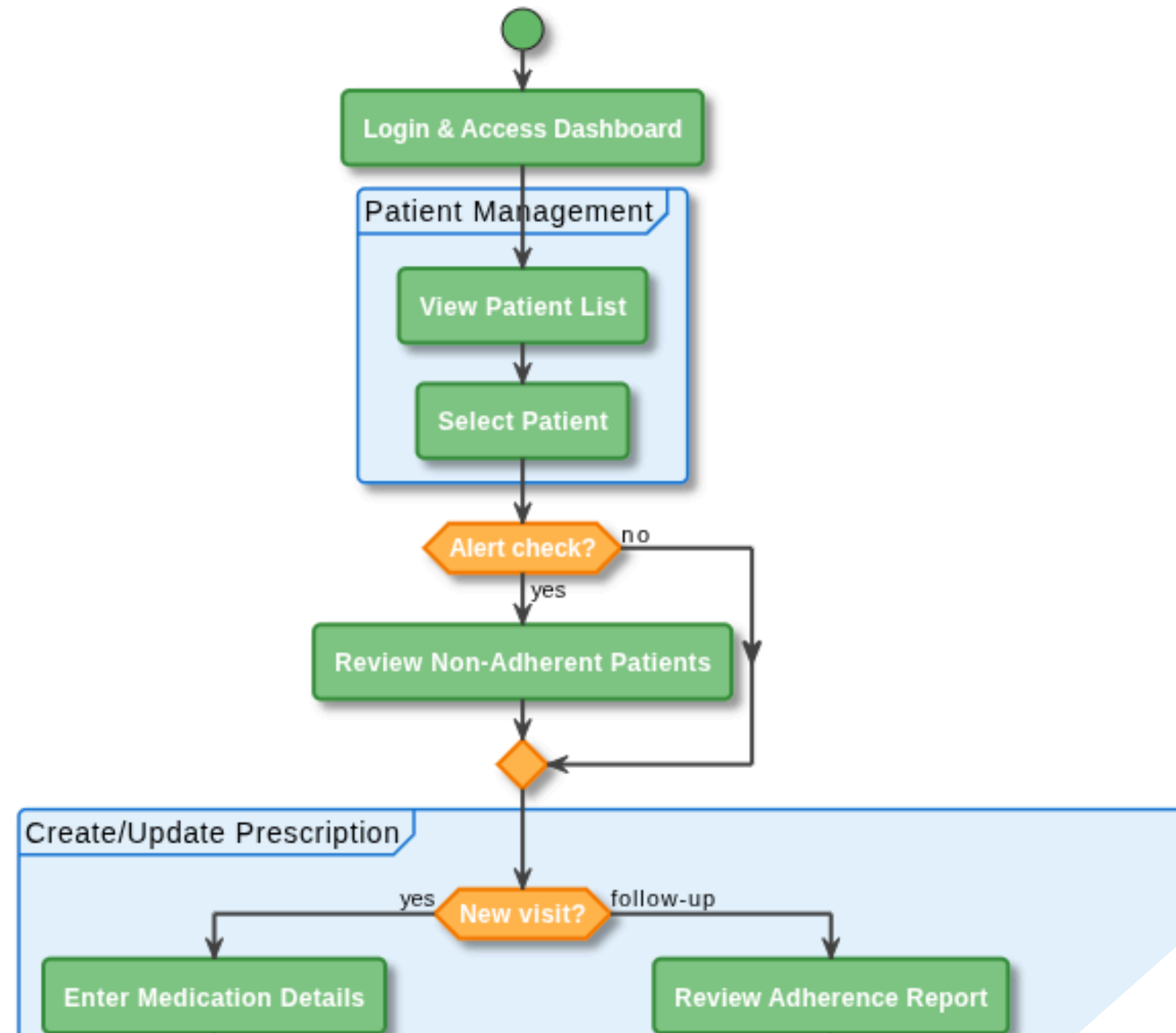


Activity Diagram (Patient)

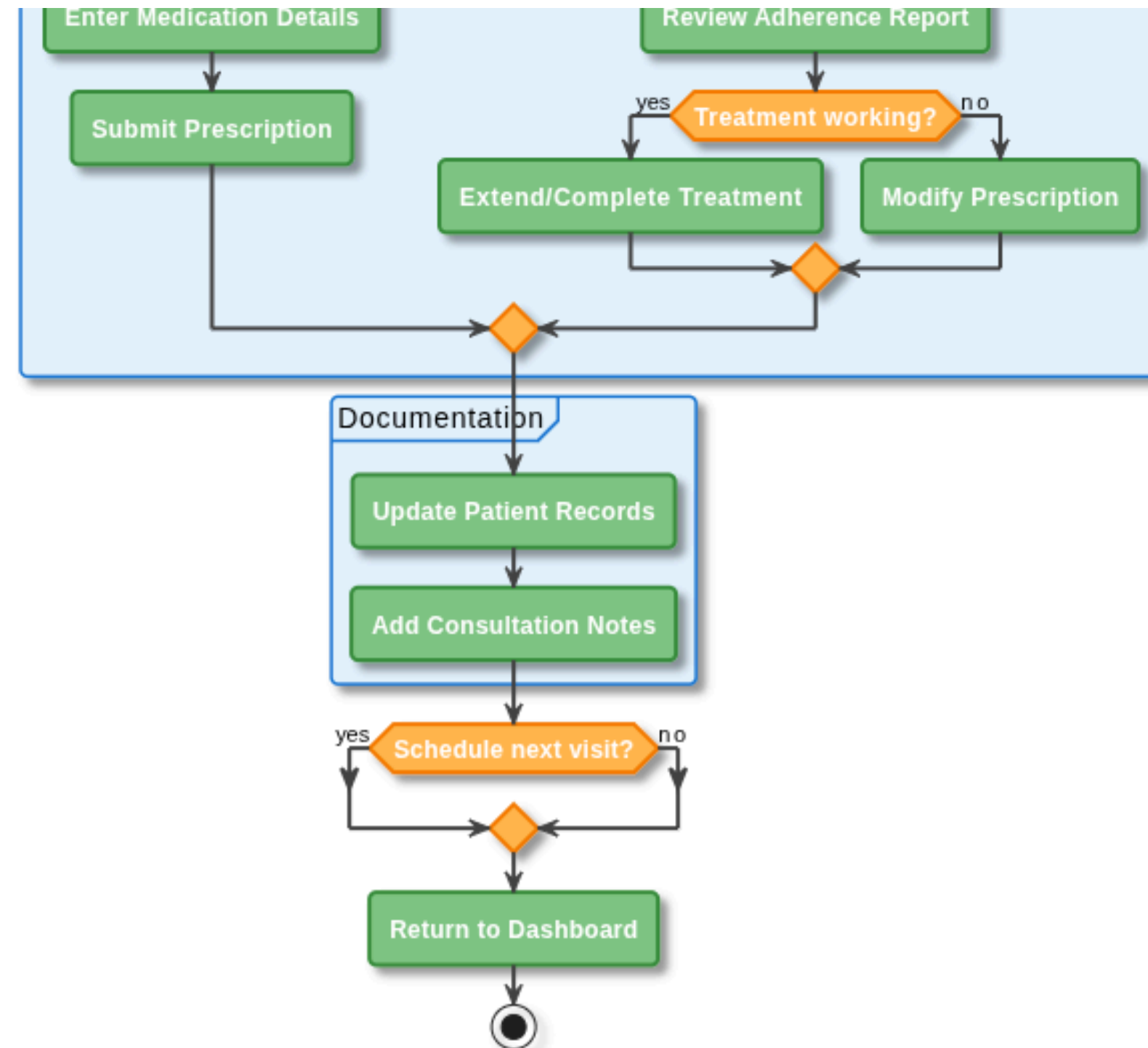


Activity Diagram (Doctor)

DasTern Doctor Activity Diagram - Simplified Workflow



Activity Diagram (Doctor)



Behind the scene

